Known but Unsayable: Decoding Defaults, Not Forgetting, Hide What Language Models Know

Anonymous ACL submission

Abstract

Language models are said to collapse onto common answers and lose the long tail of knowledge. We separate three cases: knowledge a model lacks, knowledge it holds but rarely says, and knowledge it holds but cannot say at all because the sampler gives it probability exactly zero. Top-p, top-k and min-p sampling delete low-ranked tokens before sampling, so a name loses all its probability if any one of its tokens is deleted. We give an exact, sampling-free test of this, one forward pass per answer, with a numeric margin that makes the verdict robust to floating-point differences between implementations. On 2,312 long-tail entities, 95.1% of the entities that Qwen2.5-7B-Instruct verifies (yes/no and four-way choice against near-miss foils) can never be produced under top-p 0.9, and 98.9% under its shipped sampler, against 52.4% for its base model. Across the four post-training stages of OLMo-2 the share rises from 54.4% to 93.8%, most of it at supervised fine-tuning, and the cut moves to the first token of the name. Any estimate of model knowledge built from samples therefore measures the sampler, and we compute the gap exactly. On 277 categories from 24 countries the rare entities are the ones deleted: each tenfold increase in an entity's Wikipedia coverage multiplies its odds of being unreachable by 0.15. From samples under the shipped sampler, Chao1 estimates 141 entities where the model recognises 15,514. Switching truncation off at the first token recovers only 12.3% of the unreachable entities, because names fail again later, so we report the frontier between reachable knowledge and invented answers instead.

1 Introduction

Ask a chat model to name a poet who wrote in Tamil and it will usually say Thiruvalluvar. Ask it whether *Kumaragurupara Desikar* was a Tamil poet and it will usually say yes. Both behaviours are expected. The question this paper asks is different: can the model say *Kumaragurupara Desikar* at all when asked for a Tamil poet, however many times we ask?

Under the sampler the model ships with, often it cannot. Nucleus (top-p), top-k and min-p sampling keep only the most likely tokens at each step and delete the rest before sampling. If one token of a name falls outside the kept set at any step, the probability of the whole name is exactly zero. No number of samples, retries or self-consistency votes can produce it.

This matters for the current debate on homogenisation and knowledge collapse. When a model never produces a rare fact, three different things can be true: the model does not store it; it stores it but gives it a small probability, so enough samples will find it; or it stores it and the sampler deletes it. The third case looks exactly like the first from samples, but it is a property of a decoding setting, not of the model's knowledge.

We make the third case measurable. Our contributions:

1. **Recognised-but-unreachable knowledge.** We pair an exact reachability test (one teacher-forced pass per answer, also used for ordinary words by Awad et al., 2026) with a recognition test against near-miss foils, and add a numeric margin so the verdict does not depend on floating-point differences between two computations of the same logits. Real samples from two model families never contradict it (Section 2).

2. **How much.** Among entities Qwen2.5-7B-Instruct recognises, 98.9% of those in our pilot are unreachable under its shipped sampler (Section 4).

3. **Where it comes from.** Across OLMo-2's post-training stages RBU rises mostly at supervised fine-tuning, the cut moves to the first token of the name, and first-token concentration

tracks RBU; truncation by probability mass, not by rank, does the deleting (Section 5).

4. **What it does to measurement.** Unseen-species estimators applied to model samples converge to the reachable set, not to what the model recognises; we compute the gap exactly (Section 6).

5. **Who loses, and what a fix can and cannot do.** Rare entities are deleted more often (Section 7). Relaxing truncation at the first token alone recovers little, because names fail again later; we report the full frontier between reachable knowledge and invented answers instead (Section 8).

2 Exact reachability

Samplers. At each decoding step a sampler applies, in the order used by Hugging Face `transformers`: repetition penalty, temperature T, top-k, top-p and min-p. Each step sets some logits to $-\infty$; sampling draws from the softmax of the rest. A token is *kept* if its processed logit is finite.

Keep rules. For a target token t with logit z_t: top-k keeps t iff fewer than k tokens have a larger logit; top-p keeps t iff $G(t) < p$, where $G(t)$ is the probability, at temperature T, of the tokens strictly more likely than t; min-p keeps t iff its probability is at least *min-p* times the top token's. We prove these rules equal the library's processors in unit tests on random logits.

Reachability of an answer. Write an entity e after prompt x as tokens $t_1 \dots t_L$ (the tokenisation the model itself produces in context). e is reachable under sampler S iff every t_j is kept given x and $t_{<j}$. One teacher-forced pass gives every t_j's logits, so reachability is exact and needs no sampling. It also yields critical values: $p^*(e) = \max_j G_j(e)$ (e is reachable under top-p iff $p > p^*$), $k^*(e)$ and the critical min-p. For combined samplers (the shipped settings) we run the library's own processor chain at every position. An entity is reachable if any of its surface forms (label or up to three aliases) is.

A numeric margin. The same logits computed two ways (a batched teacher-forced pass, and key-value-cached generation) differ slightly in bfloat16, and bfloat16 logits often tie. A token on a truncation boundary can therefore be kept by one computation and cut by the other. We call e *robustly unreachable* if some token stays cut even after its logit is raised by ϵ relative to every other token. Every processor above is monotone in this shift, and the shift breaks all ties. ϵ is calibrated per model before each experiment: we sample about two thousand answers from the model, compute for each the logit shift at which its verdict would flip, and take the smallest value on a fixed grid that covers every sampled answer (Appendix A). For Qwen2.5-7B-Instruct, $\epsilon = 0.75$; the largest relative logit gap among the tokens that decide a verdict was 0.94 logits. Headline numbers use the robust verdict, a lower bound; exact and upper-bound values are reported beside them (Appendix efapp:margin).

Validation. We draw 1,000 real samples per category from `generate()` under each claimed sampler and check that no answer we call robustly unreachable is ever produced. Across 22,000 samples for Qwen2.5-7B-Instruct and 33,000 for the OLMo-2 post-trained stages there were 0 and 0 violations. On CPU in float32 the two computations agree exactly and every sampled answer is scored reachable.

3 TailCat and recognition

Categories. TailCat pairs categories of long-tail entities with near-miss foils from sibling categories, built from Wikidata: for example poets who wrote in Tamil (foils: Telugu and Kannada poets) or rivers in Kerala (foils: rivers in Karnataka and Tamil Nadu). Members are stratified over popularity (Wikipedia sitelinks) and must have articles in at least two languages, which removes single-wiki bot imports whose transliterated labels no reader could know. The pilot has 11 categories (2,312 entities) including a head control, US state capitals. The full TailCat crosses 14 entity types (people, places, works and organisations) with 24 countries, 17 of them in South Asia, Africa, Southeast Asia and Latin America, and adds writers in 12 Indian languages. Foils come from the same type in neighbouring or easily confused countries. We keep a category only if it yields at least 30 members and 30 foils: 277 categories and 33,437 sampled members (at most 150 per category).

Recognition. An entity counts as recognised if the model (i) answers "Yes" to 'Is "e" *category*?' with a score $V(e) = P(\text{Yes})/(P(\text{Yes}) + P(\text{No}))$ at or above the 95th percentile of the category's foils, so at most 5% of foils pass, and (ii) picks e out of three foils in a four-way choice. RBU is the

	pilot	full TailCat
Qwen2.5-7B-Instruct, shipped	98.9%	98.8%
Qwen2.5-7B-Instruct, top-p 0.9	95.1%	91.9%
Qwen2.5-7B base, top-p 0.9	52.4%	55.0%
OLMo-2-Instruct, top-p 0.9	93.8%	91.2%
OLMo-2 base, top-p 0.9	54.4%	62.7%

Table 1: RBU: share of recognised entities that are robustly unreachable (mean over three prompt wordings). OLMo-2 ships no sampler (greedy decoding).

share of recognised entities that are unreachable. The yes/no probe separates members from foils with AUROC 0.835 for Qwen2.5-7B-Instruct on the pilot and 0.847 on the full TailCat, below the 0.85 we set in advance; the main results hold on the categories where it exceeds 0.85 on its own, and under a second probe wording (Appendix B).

4 How much recognised knowledge is unreachable?

Table 1 reports RBU for the two families. Under the sampler Qwen2.5-7B-Instruct ships with (temperature 0.7, top-k 20, top-p 0.8, repetition penalty 1.05) 98.9% of the recognised pilot entities are unreachable (95% cluster-bootstrap CI 97.4% to 99.6%). Under plain top-p 0.9 it is 95.1% (87.8% to 98.1%), against 52.4% for the base model. On the full TailCat the numbers are 91.9% (90.4% to 93.4%) for Qwen2.5-7B-Instruct and 91.2% for OLMo-2-Instruct, against 55.0% and 62.7% for their base models.

The effect is not a quirk of wording or temperature. Across five prompt wordings the coefficient of variation of RBU at top-p 0.9 is 0.019 for Qwen2.5-7B-Instruct and 0.009 for OLMo-2-Instruct. Lowering the temperature raises RBU: for the Qwen base model from 52.8% at $T = 1$ to 85.2% at $T = 0.7$ and 92.0% at $T = 0.5$. Counting only the canonical label, not aliases, gives 94.8% against 94.1% with aliases.

Scale does not remove it. Across Qwen2.5-Instruct at 0.5B, 1.5B, 3B and 7B parameters the number of recognised pilot entities grows (328, 868, 878 and 1,310), and the share unreachable under each model's shipped sampler stays high (79.4%, 98.3%, 96.4% and 98.9%). The smallest model's recognition test is weak (AUROC 0.64), so its value is the least reliable.

Even head knowledge collapses. The head control, US state capitals, was meant to be reachable by every model. The base model reaches all 49 capitals in our pilot list under top-p 0.9; the instruct model reaches 8.

5 Where it comes from

Post-training, stage by stage. OLMo-2 releases its base model and each post-training stage. RBU at top-p 0.9 is 54.4% for the base model, 89.6% after supervised fine-tuning, 93.8% after preference tuning and 93.8% for the final model. Most of the rise happens at supervised fine-tuning. Among entities recognised by both, 348 are unreachable only for the final model and 3 only for the base model; for Qwen2.5 the counts are 456 and 5 (sign tests far below $p = 0.01$). Neither fine-tuning objective can set a probability to zero by itself: the KL-regularised preference optimum keeps the support of the reference model, and cross-entropy training keeps logits finite. What they change is where the mass goes.

The first token. The share of unreachable recognised entities whose first failing token is the first token of the name rises from 32.4% (OLMo-2 base) to 73.8% (OLMo-2-Instruct), and from 28.0% to 84.1% for Qwen2.5. The mean probability of the single most likely first answer token rises with it, from 0.25 to 0.53 (OLMo-2) and from 0.19 to 0.52 (Qwen2.5). Across 66 (model, category) points, first-token concentration and RBU have Spearman correlation 0.82. Appendix C shows real names token by token: the instruct model rules them out at the first token, while the base model starts them and loses them later.

Mass, not rank. Top-k RBU barely moves across OLMo-2's stages (46.8% to 49.1% at $k = 50$) while top-p RBU rises steeply. Post-training changes how much probability the head takes far more than the order of the candidates, so truncation by mass (top-p, min-p) is what deletes the tail.

6 What it does to measurement

Let $q_S(e)$ be the probability that one sample under sampler S is entity e. After N samples the expected number of distinct entities seen is $\sum_e [1 - (1 - q_S(e))^N]$, which rises to the size of the reachable set $\{e : q_S(e) > 0\}$. Every unseen-species estimator, from Good-Turing to Chao1 and ACE, estimates the size of that support. Applied to truncated samples it estimates the reachable set, never the recognised set, and the shortfall is

$\mathrm{RBU}(S) \times |\text{recognised}|$, which our test computes exactly. Li et al. (2025) observed that nucleus sampling lowers their knowledge estimates and sampled without it; the argument above explains why and quantifies it.

Empirically, on the full TailCat with 1,000 samples per category under Qwen2.5-7B-Instruct's shipped sampler, the samples name 141 distinct members; Chao1 estimates 141 and ACE 141, while 231 members are reachable and the model recognises 15,514. Only 6 sampled members lie outside the computed reachable set, each produced through a non-canonical token path (Limitations); no sample matched the token sequence of a robustly unreachable answer. Under top-p 0.9 the corresponding numbers are 432, 470, 2,134 and 15,514.

7 Who loses

Popularity. We fit a logistic model of unreachability on $\log_{10}(1+\text{sitelinks})$ for recognised entities, with generalised estimating equations clustered by category (Liang and Zeger, 1986). For Qwen2.5-7B-Instruct each tenfold increase in Wikipedia language coverage multiplies the odds of being unreachable by 0.15 (95% CI 0.12 to 0.18); for OLMo-2-Instruct by 0.10. The same model for recognition among all members gives 2.14 for Qwen2.5-7B-Instruct.

Regions and types. Appendix D breaks RBU down by world region and entity type. RBU is not higher for categories from South Asia, Africa or Southeast Asia than for European or North American ones; it is lower. A likely reason, which we have not tested, is composition: the instruct model recognises more members of Western categories, and a category's many recognised members compete for the same few first tokens, so a larger share of them is cut. The robust pattern is within categories: the rarer a recognised entity, the more likely it is unreachable.

8 A targeted fix

If most failures happen at the first token, a natural fix is to switch truncation off for the first answer token only and keep the shipped sampler afterwards. We tested this against a sweep of standard samplers on the pilot (1,000 samples per category per sampler), measuring RBU, the number of distinct real members produced and the invented-name rate (answers that match no member of the category in Wikidata and no foil).

The fix fails our pre-registered bar of recovering half of the unreachable recognised entities. It recovers 12.3% for Qwen2.5-7B-Instruct (RBU from 98.9% to 86.7%, distinct members sampled from 15 to 41, invented-name rate -0.8 points) and 23.7% for OLMo-2-Instruct (invented-name rate +3.1 points). The reason is instructive. The first failing token is usually the first token, but most names fail again later under the same sampler, so relaxing one position rescues only names whose every failure is at the start.

The full frontier is the more useful result (Appendix E). Removing truncation entirely makes every recognised entity reachable at a cost of 56.7% invented answers against 52.4% under the shipped sampler for Qwen2.5-7B-Instruct; top-p 0.99, top-k 100 and epsilon sampling lie in between. The automatic invented-name rate overstates invention: it counts real members spelled differently or missing from Wikidata, and wrong answers, as invented. With looser matching the rates fall by about ten points while the differences between samplers barely change (relaxation: +0.5 and +2.3 points). A hand audit of 200 such answers is pending.

9 Related work

Truncation and what it removes. Nucleus sampling (Holtzman et al., 2020), its desmoothing view with epsilon and eta sampling (Hewitt et al., 2022) and min-p (Nguyen et al., 2025) all trade the tail of the distribution for fluency. Lee et al. (2022) show top-p can harm factuality and adapt it. The closest measurement is Awad et al. (2026), who use the same token-survival test to ask which *words* of book continuations remain reachable, compare base and aligned models and tabulate shipped defaults; Garces Arias et al. (2026) show that human-chosen tokens often fall outside common truncation sets. We ask a different question: whether *knowledge the model demonstrably recognises* is reachable, why post-training changes that (stage by stage, through the first answer token), and what follows for knowledge estimates. We also make the verdict robust to numeric differences between implementations, which the per-token test needs at truncation boundaries.

Knowing without saying. Gekhman et al. (2025) find that a model can rank the gold answer first internally and never generate it in 1,000 samples

drawn at temperature 1, without analysing truncation; Calderon et al. (2026) find that recall, not encoding, is the bottleneck for parametric factuality. Long-tail knowledge is harder to learn (Kandpal et al., 2023) and to trust (Mallen et al., 2023); a recent survey names decoding-time truncation as a possible cause without measuring it (Badhe et al., 2026). Our case is the zero-probability one: no amount of sampling helps.

Homogenisation and post-training. Post-training reduces output diversity (Kirk et al., 2024); open-ended answers converge within and across models (Jiang et al., 2025) and across languages (Davey et al., 2026); diversity collapses at identifiable post-training stages (Karouzos et al., 2026); and prompting can recover suppressed diversity (Zhang et al., 2026). Kalavasis et al. (2025) prove a trade-off between breadth and hallucination. These works measure diversity at a fixed decoding setting; Jiang et al. (2025) find homogeneity persists under min-p. We separate the part of collapse that is a decoding artefact, exactly.

Estimating what a model knows from samples. Li et al. (2025) estimate unseen knowledge with smoothed Good-Turing and observe that nucleus sampling lowers their estimates, so they sample without it. We show why any such estimator converges to the reachable set under truncation and compute the gap exactly, using Chao1 (Chao, 1984) and ACE (Chao and Lee, 1992) as examples.

Limitations

Recognition. Our recognition test is a proxy. Its yes/no part separates members from near-miss foils with AUROC around 0.8, below the 0.85 we set in advance; chat models answer "Yes" readily to plausible foils. The four-way choice and the per-category foil threshold limit false positives, and the main results hold on the categories that pass on their own and under a second probe wording, but RBU is defined relative to this test.

Tokenisation. We test the tokenisation the model itself produces in context. A name can sometimes be produced through another token path. In the full-TailCat sample check of Qwen2.5-7B-Instruct, only 10 distinct entities (shipped sampler) and 34 (top-p 0.9) were ever produced solely through a non-canonical path, so canonical scoring overstates RBU by a negligible amount.

Numerics. Exact zero depends on how logits are computed. Our robust verdict holds for any computation within a calibrated margin of ours; implementations further away (other GPUs, quantisation) need their own calibration.

Scope. Two model families at 7B scale, English prompts and Wikidata categories. Llama-3.1 and Gemma-2 were not run, and the multilingual analysis planned for Hindi, Tamil and Bengali prompts is future work. Wikidata membership is incomplete, which affects the invented-name rate more than RBU.

References

Samer Awad, Javier Conde, Carlos Arriaga, Tairan Fu, Javier Coronado-Blázquez, and Pedro Reviriego. 2026. Lost in sampling: Assessing lexical reachability in LLMs via the word coverage score (WCS). *Preprint*, arXiv:2605.27268. Accepted to Findings of EMNLP 2026 (arXiv comment).

Sanket Badhe, Deep Shah, and Nehal Kathrotia. 2026. Long-tail knowledge in large language models: Taxonomy, mechanisms, interventions and implications. *Preprint*, arXiv:2602.16201.

Nitay Calderon, Eyal Ben-David, Zorik Gekhman, Eran Ofek, and Gal Yona. 2026. Empty shelves or lost keys? recall is the bottleneck for parametric factuality. *Preprint*, arXiv:2602.14080.

Anne Chao. 1984. Nonparametric estimation of the number of classes in a population. *Scandinavian Journal of Statistics*, 11(4):265–270.

Anne Chao and Shen-Ming Lee. 1992. Estimating the number of classes via sample coverage. *Journal of the American Statistical Association*, 87(417):210–217.

Xaver Raphael Davey, Shubhra Mishra, Anton Adelöw, and Siyao Liu. 2026. The multilingual artificial hivemind: Do all LLMs think alike in every language? *Preprint*, arXiv:2610.16755.

Esteban Garces Arias, Nurzhan Sapargali, Christian Heumann, and Matthias Aßenmacher. 2026. The truncation blind spot: How decoding strategies systematically exclude human-like token choices. *Preprint*, arXiv:2603.18482. Accepted at INLG 2026 (arXiv comment).

Zorik Gekhman, Eyal Ben David, Hadas Orgad, Eran Ofek, Yonatan Belinkov, Idan Szpektor, Jonathan Herzig, and Roi Reichart. 2025. Inside-out: Hidden factual knowledge in LLMs. In *Conference on Language Modeling (COLM 2025)*.

John Hewitt, Christopher D. Manning, and Percy Liang. 2022. Truncation sampling as language model

desmoothing. In *Findings of the Association for Computational Linguistics: EMNLP 2022*, pages 3414–3427, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.

Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes, and Yejin Choi. 2020. The curious case of neural text degeneration. In *International Conference on Learning Representations (ICLR 2020)*.

Liwei Jiang, Yuanjun Chai, Margaret Li, Mickel Liu, Raymond Fok, Nouha Dziri, Yulia Tsvetkov, Maarten Sap, Alon Albalak, and Yejin Choi. 2025. Artificial hivemind: The open-ended homogeneity of language models (and beyond). In *NeurIPS 2025 Datasets and Benchmarks Track*. Best Paper (Datasets and Benchmarks track). Author list follows arXiv v2 (camera-ready); the NeurIPS site lists nine authors, without Alon Albalak.

Alkis Kalavasis, Anay Mehrotra, and Grigoris Velegkas. 2025. On the limits of language generation: Trade-offs between hallucination and mode collapse. In *57th Symposium on Theory of Computing (STOC 2025)*.

Nikhil Kandpal, Haikang Deng, Adam Roberts, Eric Wallace, and Colin Raffel. 2023. Large language models struggle to learn long-tail knowledge. In *International Conference on Machine Learning (ICML 2023)*.

Constantinos Karouzos, Xingwei Tan, and Nikolaos Aletras. 2026. Where does output diversity collapse in post-training? *Preprint*, arXiv:2604.16027.

Robert Kirk, Ishita Mediratta, Christoforos Nalmpantis, Jelena Luketina, Eric Hambro, Edward Grefenstette, and Roberta Raileanu. 2024. Understanding the effects of RLHF on LLM generalisation and diversity. In *International Conference on Learning Representations (ICLR 2024)*.

Nayeon Lee, Wei Ping, Peng Xu, Mostofa Patwary, Pascale Fung, Mohammad Shoeybi, and Bryan Catanzaro. 2022. Factuality enhanced language models for open-ended text generation. In *NeurIPS 2022*.

Xiang Li, Jiayi Xin, Qi Long, and Weijie J. Su. 2025. Evaluating the unseen capabilities: How many theorems do LLMs know? *Preprint*, arXiv:2506.02058.

Kung-Yee Liang and Scott L. Zeger. 1986. Longitudinal data analysis using generalized linear models. *Biometrika*, 73(1):13–22.

Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Daniel Khashabi, and Hannaneh Hajishirzi. 2023. When not to trust language models: Investigating effectiveness of parametric and non-parametric memories. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9802–9822, Toronto, Canada. Association for Computational Linguistics.

Minh Nhat Nguyen, Andrew Baker, Clement Neo, Allen Roush, Andreas Kirsch, and Ravid Shwartz-Ziv. 2025. Turning up the heat: Min-p sampling for creative and coherent LLM outputs. In *Proceedings of the 2025 International Conference on Learning Representations (ICLR)*.

Jiayi Zhang, Simon Yu, Derek Chong, Anthony Sicilia, Michael R. Tomz, Christopher D. Manning, and Weiyan Shi. 2026. Verbalized sampling: How to mitigate mode collapse and unlock LLM diversity. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pages 160073–160152.

A Numeric margin

For each model we sample 500 answers per pilot category with a separate seed, compute for each distinct answer the logit shift at which its reachability verdict flips (by bisection), and set ϵ to the smallest value on the grid $\{0.25, 0.5, 0.75, 1, 1.5, 2, 3, 4, 6\}$ that covers every sampled answer. For Qwen2.5-7B-Instruct, 1,685 distinct sampled answers gave 63 exact-verdict contradictions and none at $\epsilon = 0.75$; the 99.9th percentile of the relevant logit gap was 0.72. In the Qwen sample check, 1 sampled answer would have contradicted the exact verdict and none contradicted the robust one.

B Recognition validity

category	model	AUROC	4-way acc.	recognised	foils passing
assamese_writers	instruct	0.716	0.711	18	5.0
bengali_film_directors	instruct	0.810	0.888	177	5.0
brazil_universities	instruct	0.979	0.835	177	5.1
finland_lakes	instruct	0.881	0.818	280	5.0
japanese_novelists	instruct	0.987	0.978	384	5.0
kerala_rivers	instruct	0.806	0.500	35	5.6
nepal_mountains	instruct	0.776	0.550	50	5.5
nigerian_painters	instruct	0.891	0.644	31	6.7
tamil_poets	instruct	0.802	0.740	58	5.4
tamilnadu_hindu_temples	instruct	0.738	0.555	73	5.0
us_state_capitals	instruct	0.874	0.939	27	6.8
ALL (pooled)	instruct	0.835	0.767	1310	–
assamese_writers	base	0.677	0.651	21	5.0
bengali_film_directors	base	0.833	0.819	153	5.0
brazil_universities	base	0.978	0.839	179	5.1
finland_lakes	base	0.799	0.845	244	5.0
japanese_novelists	base	0.987	0.873	346	5.0
kerala_rivers	base	0.695	0.510	23	5.6
nepal_mountains	base	0.746	0.544	51	5.5
nigerian_painters	base	0.824	0.630	22	6.7
tamil_poets	base	0.856	0.699	55	5.4
tamilnadu_hindu_temples	base	0.758	0.482	83	5.0
us_state_capitals	base	0.948	0.939	38	6.8
ALL (pooled)	base	0.814	0.728	1215	–

Table 2: Recognition validity on the pilot (Qwen2.5-7B pair, a-priori probe). Source: results/G0_5090/metrics/validity_*.csv.

C Walk-through examples

Each table shows one recognised entity token by token under top-p 0.9 (template 0): the chat model's verdict first, then the base model's. Source: extttresults/*/examples/.

Shintarō Ishihara (japanese_novelists, Qwen2.5-7B)

model	token	mass ahead (G, $T = 1$) / kept at $p = 0.9$
chat	'Sh'	0.998 / cut
	'int'	0.952 / cut
	'ar'	0.970 / cut
	'ō'	0.000 / kept
	' Ish'	0.000 / kept
	'ih'	0.000 / kept
	'ara'	0.000 / kept
base	' Sh'	0.799 / kept
	'int'	0.819 / kept
	'ar'	0.971 / cut
	'ō'	0.000 / kept
	' Ish'	0.000 / kept
	'ih'	0.000 / kept
	'ara'	0.000 / kept

Himalchuli (nepal_mountains, Qwen2.5-7B)

model	token	mass ahead (G, $T = 1$) / kept at $p = 0.9$
chat	'H'	0.999 / cut
	'imal'	0.000 / kept
	'ch'	0.000 / kept
	'uli'	0.000 / kept
base	' Himal'	0.847 / kept
	'ch'	0.936 / cut
	'uli'	0.000 / kept

Federal University of Science and Innovation of Minas Gerais (brazil_universities, OLMo-2-7B)

model	token	mass ahead (G, $T = 1$) / kept at $p = 0.9$
chat	'Federal'	0.914 / cut
	' University'	0.000 / kept
	' of'	0.000 / kept
	' Science'	1.000 / cut
	' and'	0.000 / kept
	' Innovation'	1.000 / cut
	' of'	0.998 / cut
	' Min'	0.887 / kept
	'as'	0.000 / kept
	' G'	0.000 / kept
	'erais'	0.000 / kept
base	' Federal'	0.261 / kept
	' University'	0.000 / kept
	' of'	0.000 / kept
	' Science'	0.995 / cut
	' and'	0.000 / kept
	' Innovation'	1.000 / cut
	' of'	0.922 / cut
	' Min'	0.410 / kept
	'as'	0.000 / kept
	' G'	0.000 / kept
	'erais'	0.000 / kept

State University of Feira de Santana (brazil_universities, OLMo-2-7B)

model	token	mass ahead (G, $T = 1$) / kept at $p = 0.9$
chat	'State'	0.993 / cut
	' University'	0.000 / kept
	' of'	0.000 / kept
	' Fe'	1.000 / cut
	'ira'	0.000 / kept
	' de'	0.000 / kept
	' Santana'	0.000 / kept
base	' State'	0.777 / kept
	' University'	0.000 / kept
	' of'	0.000 / kept
	' Fe'	0.915 / cut
	'ira'	0.000 / kept
	' de'	0.000 / kept
	' Santana'	0.000 / kept

D Breakdowns

region	OLMo-2-1124-7B	OLMo-2-1124-7B-Inst	Qwen2.5-7B	Qwen2.5-7B-Inst
africa	49.7	82.5	42.5	85.8
east_asia	70.1	97.5	65.2	98.9
europe	76.6	95.8	69.1	96.8
latin_america	60.4	91.5	56.6	91.5
north_america	79.7	99.0	65.6	99.5
south_asia	49.7	83.9	43.0	87.3
southeast_asia	53.0	86.1	42.7	87.1
west_asia	61.4	91.4	53.6	90.1

Table 3: Robust RBU (%) at top-p 0.9 by region, full TailCat. Source: results/P3/g2/by_region.csv.

type	OLMo-2-1124-7B	OLMo-2-1124-7B-Inst	Qwen2.5-7B	Qwen2.5-7B-Inst
architects	64.8	88.6	53.2	87.1
composers	54.6	87.6	48.5	86.4
film_directors	57.8	91.6	50.8	88.4
films	44.9	84.8	53.6	93.1
football_clubs	65.6	93.2	71.4	96.4
islands	72.4	91.4	58.4	89.3
lakes	73.3	96.1	64.2	94.4
mountains	77.5	95.8	63.1	95.1
museums	69.7	97.2	66.0	98.0
novelists	59.6	88.4	54.5	92.4
painters	54.4	85.4	42.6	82.5
poets	57.8	90.4	51.0	87.4
rivers	69.4	91.2	49.7	96.8
universities	62.4	96.8	50.2	99.1
writers_by_language	39.7	79.7	38.9	82.5

Table 4: Robust RBU (%) at top-p 0.9 by type, full TailCat. Source: results/P3/g2/by_type.csv.

E Mitigation frontier

F Reproducibility

model	sampler	RBU	recovered	distinct members	invented	Δ invented (pts)
Qwen2.5-7B-Instruct	untruncated_T1	0.0	100.0	75	56.7	+4.3
Qwen2.5-7B-Instruct	p0.80_T1	97.7	1.2	30	54.2	+1.8
Qwen2.5-7B-Instruct	p0.90_T1	95.3	3.6	42	54.9	+2.5
Qwen2.5-7B-Instruct	p0.95_T1	91.1	7.9	55	55.4	+2.9
Qwen2.5-7B-Instruct	p0.99_T1	74.0	25.2	68	56.1	+3.7
Qwen2.5-7B-Instruct	k20_T1	79.6	19.5	71	54.3	+1.9
Qwen2.5-7B-Instruct	k50_T1	62.2	37.1	74	55.3	+2.9
Qwen2.5-7B-Instruct	k100_T1	46.6	52.8	75	55.6	+3.2
Qwen2.5-7B-Instruct	minp0.02_T1	95.0	3.9	49	53.6	+1.2
Qwen2.5-7B-Instruct	minp0.05_T1	97.2	1.7	31	52.8	+0.4
Qwen2.5-7B-Instruct	minp0.10_T1	98.2	0.6	28	51.8	-0.6
Qwen2.5-7B-Instruct	eps3e-4_T1	75.9	23.2	74	55.5	+3.1
Qwen2.5-7B-Instruct	reference	98.9	0.0	15	52.4	+0.0
Qwen2.5-7B-Instruct	relax_first	86.7	12.3	41	51.6	-0.8
OLMo-2-1124-7B-Instruct	untruncated_T1	0.0	100.0	78	47.0	+4.3
OLMo-2-1124-7B-Instruct	p0.80_T1	97.2	-3.3	25	41.9	-0.8
OLMo-2-1124-7B-Instruct	p0.90_T1	94.0	0.0	36	42.7	+0.0
OLMo-2-1124-7B-Instruct	p0.95_T1	89.8	4.5	63	43.7	+0.9
OLMo-2-1124-7B-Instruct	p0.99_T1	58.6	37.7	77	45.8	+3.1
OLMo-2-1124-7B-Instruct	k20_T1	70.5	25.1	65	45.6	+2.9
OLMo-2-1124-7B-Instruct	k50_T1	49.6	47.3	81	46.2	+3.5
OLMo-2-1124-7B-Instruct	k100_T1	35.1	62.6	81	46.6	+3.9
OLMo-2-1124-7B-Instruct	minp0.02_T1	93.9	0.2	47	43.0	+0.3
OLMo-2-1124-7B-Instruct	minp0.05_T1	96.8	-2.9	32	41.9	-0.8
OLMo-2-1124-7B-Instruct	minp0.10_T1	98.3	-4.5	18	39.9	-2.8
OLMo-2-1124-7B-Instruct	eps3e-4_T1	64.4	31.5	76	46.1	+3.4
OLMo-2-1124-7B-Instruct	reference	94.0	0.0	36	42.7	+0.0
OLMo-2-1124-7B-Instruct	relax_first	71.7	23.7	68	45.8	+3.1

Table 5: Mitigation frontier on the pilot (template 0, 1,000 samples per category per sampler). Reference: shipped sampler (Qwen2.5), top-p 0.9 (OLMo-2, which ships none). Source: results/P5.

run	model	snapshot	GPU	ϵ	minutes	code
G0_5090	Qwen2.5-7B	`d149729`	GeForce RTX 5090	0.75	2.3	`9486a4b`
G0_5090	Qwen2.5-7B-Instruct	`a09a354`	GeForce RTX 5090	0.75	4.6	`9486a4b`
G1	OLMo-2-1124-7B	`7df9a82`	GeForce RTX 5090	0.5	2.4	`7a50eb9`
G1	OLMo-2-1124-7B-DPO	`e34ea60`	GeForce RTX 5090	0.5	2.4	`7a50eb9`
G1	OLMo-2-1124-7B-Instruct	`470b1fb`	GeForce RTX 5090	0.5	2.4	`7a50eb9`
G1	OLMo-2-1124-7B-SFT	`1de02c0`	GeForce RTX 5090	0.5	2.4	`7a50eb9`
P3_robust	OLMo-2-1124-7B	`7df9a82`	GeForce RTX 5090	0.5	3.3	`ff7fce2`
P3_robust	OLMo-2-1124-7B-Instruct	`470b1fb`	GeForce RTX 5090	0.5	2.4	`ff7fce2`
P3_robust	Qwen2.5-7B	`d149729`	GeForce RTX 5090	0.75	3.5	`ff7fce2`
P3_robust	Qwen2.5-7B-Instruct	`a09a354`	GeForce RTX 5090	0.75	4.4	`ff7fce2`
P3	OLMo-2-1124-7B	`7df9a82`	GeForce RTX 5090	0.5	27.6	`3053872`
P3	OLMo-2-1124-7B-Instruct	`470b1fb`	GeForce RTX 5090	0.5	39.4	`3053872`
P3	Qwen2.5-7B	`d149729`	GeForce RTX 5090	0.75	30.0	`3053872`
P3	Qwen2.5-7B-Instruct	`a09a354`	GeForce RTX 5090	0.75	79.0	`3053872`
P3_scale	Qwen2.5-0.5B-Instruct	`7ae5576`	GeForce RTX 5090	3.0	1.7	`3053872`
P3_scale	Qwen2.5-1.5B-Instruct	`989aa79`	GeForce RTX 5090	0.75	1.8	`3053872`
P3_scale	Qwen2.5-3B-Instruct	`aa8e725`	GeForce RTX 5090	2.0	2.1	`3053872`
P5	OLMo-2-1124-7B-Instruct	`470b1fb`	GeForce RTX 5090	0.5	11.3	`3053872`
P5	Qwen2.5-7B-Instruct	`a09a354`	GeForce RTX 5090	0.75	13.6	`3053872`

Table 6: Runs behind every number: model snapshot (Hugging Face commit), GPU, numeric margin, wall time and code commit. All data, outputs, metric tables and the decision log are in the repository; total GPU time for the project 5.28 hours (logs/gpu_ledger.csv).